Laboratory Work No. 1. Linear Regression.

After doing the Laboratory work No.1, fill the review table and in *.pdf form submit it to Moodle „Laboratory Work No. 1“.

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Question No.		Leave the actual value or answer a question
General requirements		
	Provide information about files submitted to moodle:	
0.1	Report is submitted in pdf format	Yes
0.2	Scripts are submitted (*.py, *.ipynb, ...)	Yes
0.3	The laboratory report includes the following sections:	
0.3.1	Student's name, surname and laboratory work number	Yes
0.3.2	Introductory part describing the task and objectives	Yes
0.3.3	Theoretical background of the model(s) used in the work	Yes
0.3.4	Data Analysis and Linear Regression sections are provided according to the requirements below.	Yes
0.3.5	Conclusions	Yes
Dataset and data analysis		
2.	Provide information about the dataset and perform data quality analysis	
2.1	The correct dataset and target variable are used. All selected features are briefly described and correctly identified as continuous or categorical.	Yes
2.2	The following statistics are provided for continuous variables: • total number of values; • percentage of missing values; • cardinality; • minimum and maximum values; • 1st and 3rd quartiles; • mean; • median; • standard deviation.	Yes
2.3	The following statistics are provided for categorical variables: • total number of values; • percentage of missing values; • cardinality; • mode; • mode frequency; • mode percentage; • second mode; • second mode frequency; • second mode percentage.	Yes
2.4	Histograms are provided for continuous variables, and their distributions are described and commented on (all variables may be displayed or only the most relevant ones).	Yes
2.5	Main data quality issues (e.g., missing values, outliers) are identified.	Yes
2.6	It is explained how the identified data quality issues will be handled and justification for the selected methods is provided.	Yes

2.7	Relationships between variables are analyzed using **at least two different types of visualizations** (e.g., scatter plots, SPLOM, bar plots). Important relationships observed in the visualizations are identified and commented on.	Yes
2.8	A correlation matrix for continuous variables is provided. The strongest or otherwise important correlations are identified and commented on.	Yes
2.9	Final data preparation for linear regression is performed: • identified data quality issues are handled; • numerical variables are normalized; • categorical variables are converted to a numerical representation (e.g., one-hot encoding, dummy coding, other appropriate encoding methods)	Yes
2.9.1	Parameters used for data transformations (normalization, missing value handling, outlier modification, etc.) are obtained from the training data and then applied to validation/test data.	Yes
2.9.2	**Optionally**, variables are additionally transformed or new derived variables are created (e.g., binning, logarithmic transformation, ratios).	Yes
Linear regression		
3.	Provide information about the implemented linear regression model.	
3.1	Calculation of linear regression coefficients is implemented using the Least Squares method.	Yes
3.2	Calculation of linear regression coefficients is implemented using Gradient Descent or one of its modifications.	Yes
3.2.1	**Optionally**, if Gradient Descent modification is used, specify the method.	
3.3	Coefficients obtained using both implemented methods are presented (e.g., in one table) and the results are compared and commented on.	Yes
3.4	The statistical significance of regression coefficients is evaluated. Significant and non-significant variables are identified, and the model is modified if necessary.	Yes
3.5	The model is cross-validated using at least 5 folds/subsets with **one of the implemented coefficient estimation methods**, and selected error metrics are calculated.	Yes
3.5.1	Coefficient estimation method used for cross-validation.	Least Squares
3.5.2	Selected error metrics used for model evaluation.	Metrics: MSE and RMSE
3.6	K-fold cross-validation is performed and compared for the following feature sets: • only significant variables; • only the 3 most significant variables; • all available variables (if non-significant variables were identified).	Yes
3.6.1	For each feature set, the mean and standard deviation of the selected error metrics are presented in a table.	Yes
3.6.2	Results are also visualized using box plots on the same axes and differences in error and dispersion are commented on.	Yes
3.6.3	The model variants from 3.6 are evaluated on the test dataset using appropriate error metrics and error analysis visualizations. The results are compared and commented on.	Yes

3.7	A linear regression model is also created using a standard library implementation (e.g., scikit-learn), and its results are compared with the implemented model.	Yes
3.8	The selected implemented model is prepared for use with unseen data. A single record can be entered in its original format, required data transformations are applied, and the target value is predicted.	Yes